

Fundamentals of Solid State Physics

Electrons in a Periodic Potential

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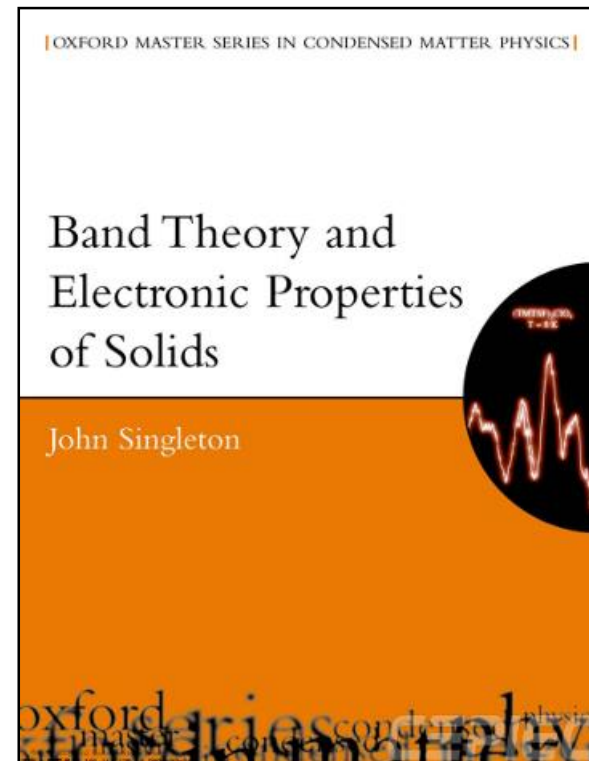
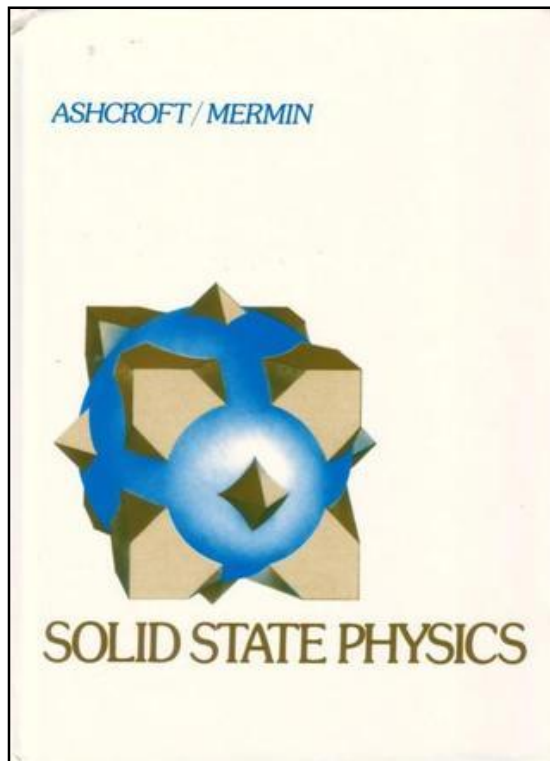


This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

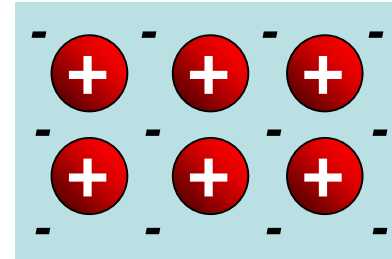
Further Reading

- Ashcroft & Mermin, Chapter 8
- Singleton, Chapter 2



The Drude Model 德鲁德模型

Free electron 'gas' model, in 1900



positive ions
+
electron cloud

- Independent
 - electrons do not interact with each other
- Free
 - electrons do not interact with ions, except collision
- Collision (Origin of the resistance)
 - electrons are scattered by the ions instantaneously
- Relaxation time τ 弛豫时间
 - average time between two collisions
 - electron mean free path $l = v^* \tau$
- Maxwell–Boltzmann distribution
 - average kinetic energy

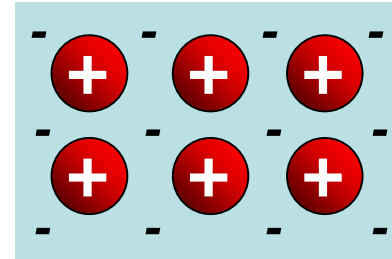


P. Drude
1863–1906

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

The Sommerfeld Model 索末菲模型

Free electron 'Fermi gas'



positive ions
+
electron cloud

- Independent
 - electrons do not interact with each other
- Free
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 - electrons are scattered by the ions instantaneously
- Relaxation time τ 弛豫时间
 - average time between two collisions
 - electron mean free path $l = v^* \tau$
- Fermi–Dirac distribution
 - quantum mechanics



A. Sommerfeld
1868–1951

The Drude and Sommerfeld Models

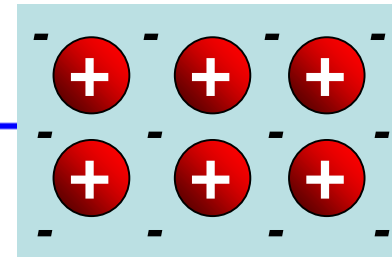
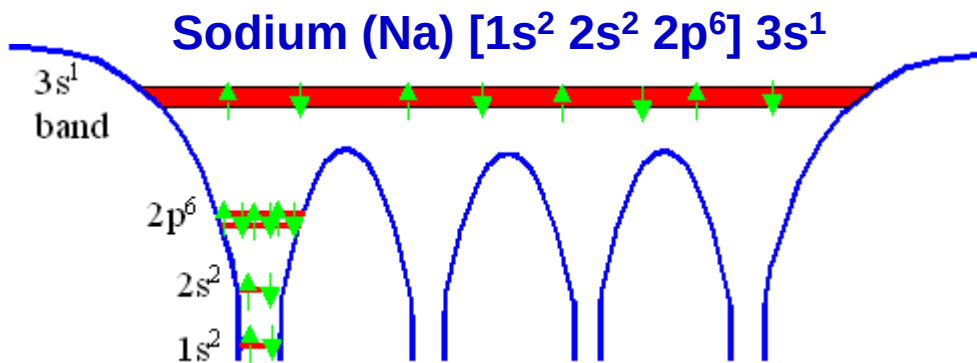
- Success for some metals
- Failure for many other cases
- They omitted:
 - ❑ Material and atom structures
 - ❑ Potentials of positive ions
 - ❑ Localized electrons
 - ❑ ...



P. Drude
1863–1906



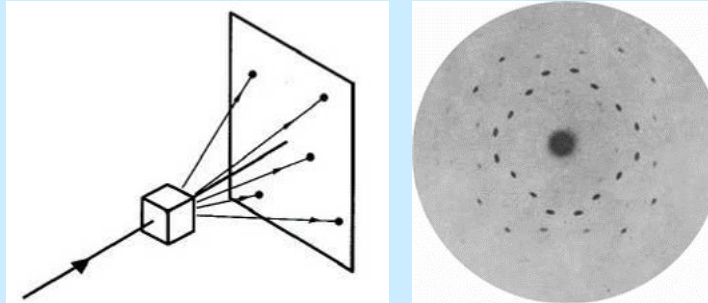
A. Sommerfeld
1868–1951



positive ions
+
electron cloud

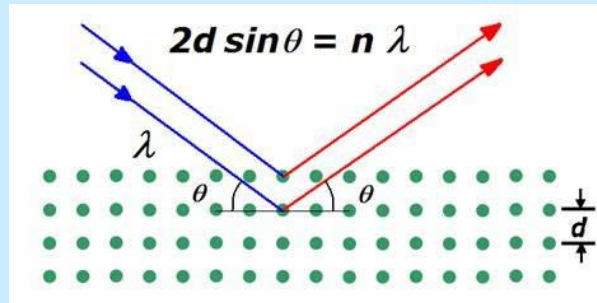
Crystal Structures

**X-ray
diffraction
of crystals**



**M. von Laue (劳厄)
Nobel Prize in 1914**

Bragg's law

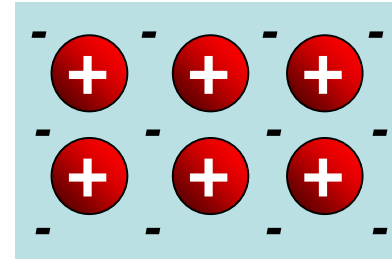


**Bragg & Bragg
(布拉格父子)
Nobel Prize in 1915**

Many solids have crystal structures:
***Periodic* atomic arrays -> *Periodic* potentials**

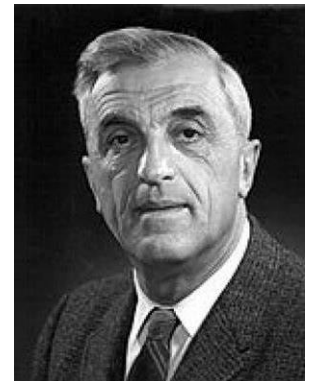
The Bloch Model 布洛赫模型

Electrons are *not free*



positive ions
+
electron cloud

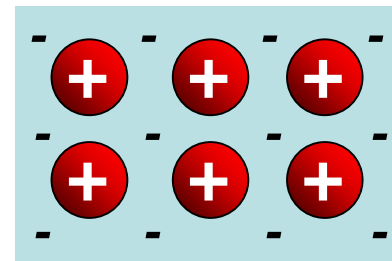
- Independent
 - electrons do not interact with each other
- Not Free
 - electrons are in a periodic potential formed by lattice
- Fermi–Dirac distribution
 - quantum mechanics



F. Bloch
1905–1983

Bloch Electrons

- Electrons are *not free*, but *in a periodic potential* formed by the atomic lattice



positive ions
+
electron cloud

- Born–Oppenheimer Approximation
 - ▣ The behaviors of electrons and nuclei can be calculated separately.
- Independent Electron Approximation
 - ▣ We still assume electrons are independent and do not interact with each other

Born–Oppenheimer Approximation

- Adiabatic Approximation 绝热近似
- Static Approximation 定核近似
 - The behaviors of electrons and nuclei can be calculated separately.

$$\Psi_{\text{total}} = \Psi_{\text{electron}} * \Psi_{\text{nuclear}}$$

- Electrons move much faster than nuclei
- When we consider the electronic behaviors, we assume the atomic lattice is static.

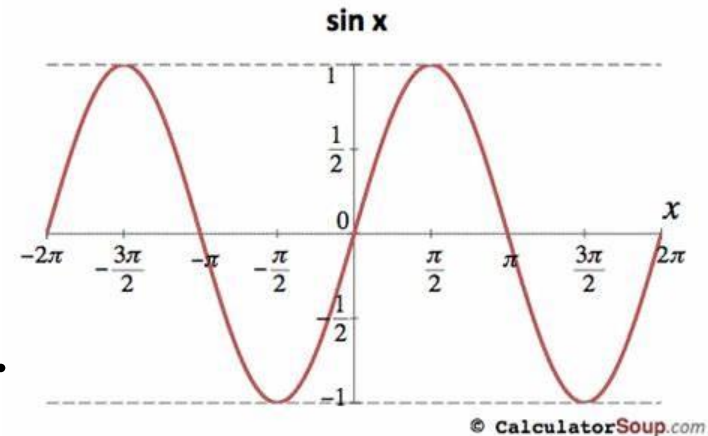


Fourier Series

- Periodic wave functions

$$f(x + na) = f(x)$$

$$n = 0, \pm 1, \pm 2, \dots$$



- We can have

$$f(x) = \sum_n A_n \cos(nkx)$$

$$n = 0, \pm 1, \pm 2, \dots$$

- or complex solutions

$$f(x) = \sum_n A_n \exp(inkx)$$

$$k = \frac{2\pi}{a}$$

A - amplitude

k - wavenumber **13**

Periodic Potentials

For a Bravais lattice

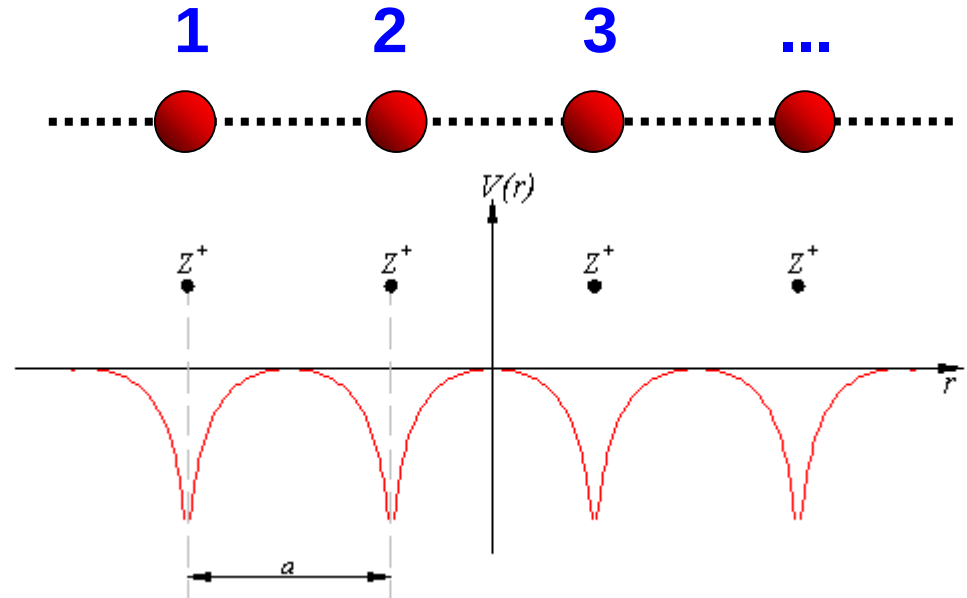
$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

$$V(\mathbf{r}) = V(\mathbf{r} + \mathbf{R})$$



$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$

→ $\exp(i\mathbf{G} \cdot \mathbf{R}) = 1$



a Fourier series

Reciprocal Lattice 倒易点阵

$$\exp(i\mathbf{G} \cdot \mathbf{R}) = 1$$

Lecture note 2.2

- For a Bravais lattice

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

n_1, n_2, n_3 are integers

- We define vector $\mathbf{G}$ as

$$\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + m_3 \mathbf{b}_3$$

m_1, m_2, m_3 are integers

$(\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ forms reciprocal lattice (倒易点阵 / 倒格子)
 $\mathbf{G}$ is in the reciprocal space (倒易空间 / 倒空间)

Reciprocal Lattice 倒易点阵

$$\exp(i\mathbf{G} \cdot \mathbf{R}) = 1 \rightarrow \mathbf{G} \cdot \mathbf{R} = 2\pi \cdot N$$

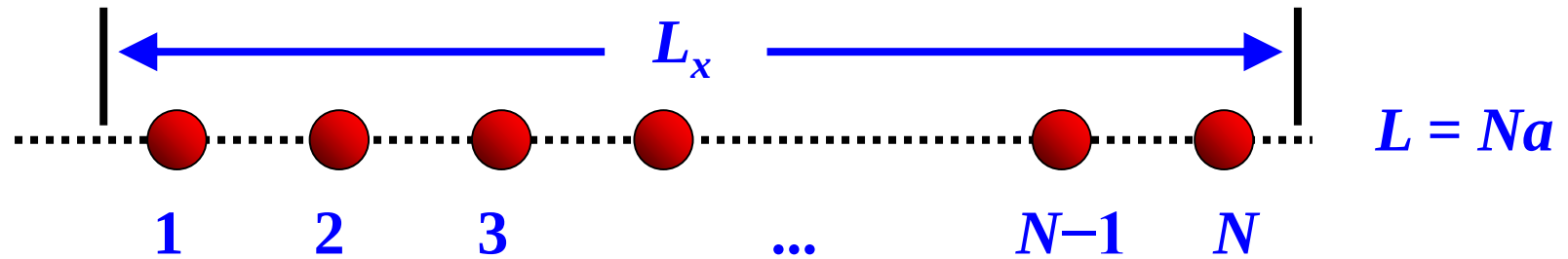
- One can have

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij} \quad \begin{cases} \delta_{ij} = 0, & i \neq j \\ \delta_{ij} = 1, & i = j \end{cases}$$

$(\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ is primitive vectors to form reciprocal lattice
(also a Bravais lattice)

The reciprocal lattice is the Fourier transform of the direct lattice

An Electron in a Periodic Potential



$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \cdot \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

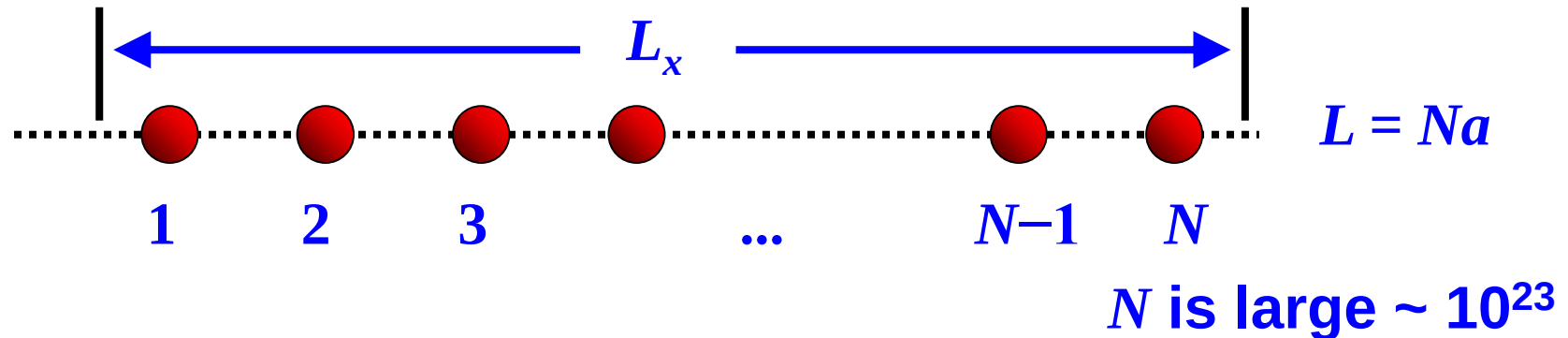
and

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$



$$\psi(\mathbf{r}) = ???$$

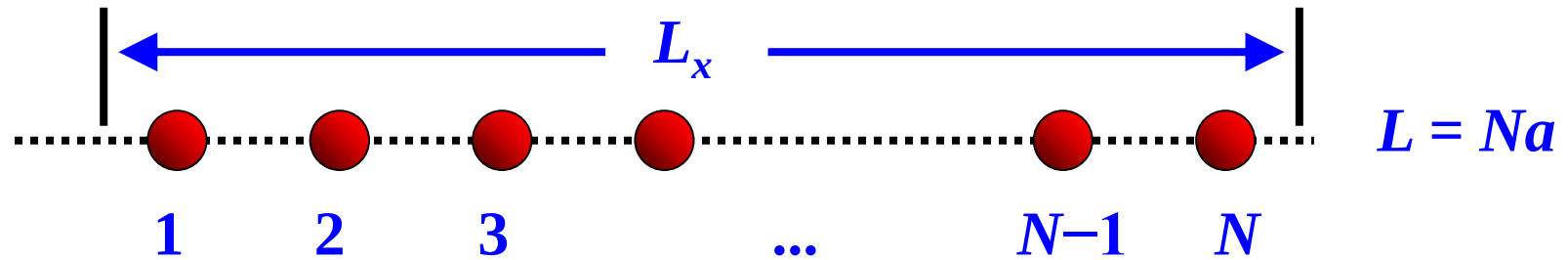
An Electron in a Periodic Potential



Born-von Karman *periodic* boundary condition

$$\psi(x) = \psi(x + L_x)$$

An Electron in a Periodic Potential



N is large $\sim 10^{23}$

$$\psi(x) = \psi(x + L_x) \rightarrow \psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r}) \quad \exp(ik_x L_x) = 1$$

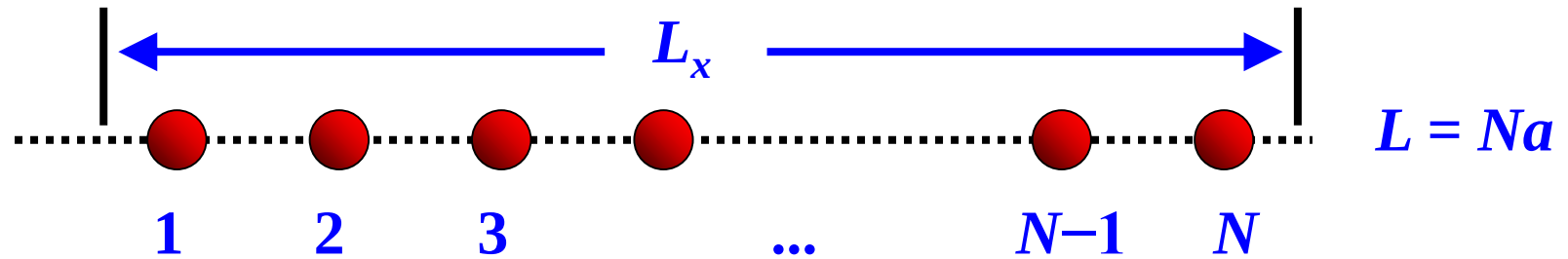
$$\rightarrow k_x = \frac{2\pi n_x}{L_x}, k_y = \frac{2\pi n_y}{L_y}, k_z = \frac{2\pi n_z}{L_z} \quad \text{quasi-continuous}$$

$n = 0, \pm 1, \pm 2, \dots$

$$L_x = n_x a_1, L_y = n_y a_2, L_z = n_z a_3$$

$$b_1 = \frac{2\pi}{a_1}, b_2 = \frac{2\pi}{a_2}, b_3 = \frac{2\pi}{a_3}$$

An Electron in a Periodic Potential



N is large $\sim 10^{23}$

$$\psi(x) = \psi(x + L_x) \rightarrow \psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r}) \quad \exp(ik_x L_x) = 1$$

$$\rightarrow k_x = \frac{2\pi n_x}{L_x}, k_y = \frac{2\pi n_y}{L_y}, k_z = \frac{2\pi n_z}{L_z} \quad \text{quasi-continuous}$$

$$\rightarrow \mathbf{k} = \frac{n_x}{N_x} \mathbf{b}_1 + \frac{n_y}{N_y} \mathbf{b}_2 + \frac{n_z}{N_z} \mathbf{b}_3 \quad n = 0, \pm 1, \pm 2, \dots$$

$\mathbf{r}, \mathbf{R}, \mathbf{k}, \mathbf{G}$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$

- $\mathbf{R}$: position of lattice point in real space
- $\mathbf{r}$: vector in real space

3D $\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$

1D $R = 0, \pm a, \pm 2a, \dots$

- $\mathbf{G}$: position of lattice point in reciprocal space
- $\mathbf{k}$: wave vector in reciprocal space

3D $\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + m_3 \mathbf{b}_3$

1D $G = 0, \pm \frac{2\pi}{a}, \pm \frac{4\pi}{a}, \dots$

An Electron in a Periodic Potential

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \cdot \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$

→

$$\left(\frac{\hbar^2 k^2}{2m} - E \right) C_{\mathbf{k}} + \sum_{\mathbf{G}} V_{\mathbf{G}} C_{\mathbf{k}-\mathbf{G}} = 0$$

The Central Equation

If we know $V(\mathbf{r})$ →

$$C_{\mathbf{k}}, C_{\mathbf{k} \pm \mathbf{G}}, C_{\mathbf{k} \pm 2\mathbf{G}} \dots$$

$$E_1, E_2, E_3 \dots$$

The Central Equation

We only need to solve it in the first Brillouin zone

$$\begin{pmatrix}
 \ddots & & & & \ddots \\
 & \ddots & & & \\
 & & \ddots & & \\
 & & & \ddots & \\
 & & & & \ddots
 \end{pmatrix}
 \begin{pmatrix}
 \vdots \\
 \vdots \\
 \vdots \\
 \vdots \\
 \vdots
 \end{pmatrix}
 =
 \begin{pmatrix}
 C_{k-g} \\
 C_{k-g} \\
 C_{k-g} \\
 C_{k-g} \\
 C_{k-g}
 \end{pmatrix}$$

The matrix is a tridiagonal matrix with elements $\frac{1}{2m}$ on the diagonal and $\frac{1}{2m}$ on the off-diagonals. The vector on the right is C_{k-g} .

$$g = \frac{2\pi}{a}$$

The Central Equation

We only need to solve it in the first Brillouin zone

$$\begin{pmatrix} \ddots & & \dots & & \dots & & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & & V_{-g} & & V_{-2g} & \dots \\ \dots & & & & & & \\ \dots & & V_g & & \frac{\hbar^2}{2m}k^2 - E & & \dots \\ \dots & & & & & & \\ \dots & & V_{2g} & & V_g & & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & & \dots & & \dots & & \ddots \end{pmatrix} \begin{pmatrix} \vdots \\ C_{k-g} \\ C_k \\ C_{k+g} \\ \vdots \end{pmatrix} = 0$$

$V_{G=0}$ is a constant (the ground level energy)
here we set $V_{G=0} = V_0 = 0$

$$g = \frac{2\pi}{a}$$

The Central Equation

We only need to solve it in the first Brillouin zone

$$\det \begin{pmatrix} \ddots & & & & \ddots \\ & \frac{\hbar^2}{2m}(k-g)^2 - E & V_{-g} & V_{-2g} & \\ \dots & & \frac{\hbar^2}{2m}k^2 - E & V_{-g} & \\ & V_g & & & \ddots \\ \dots & V_{2g} & V_g & \frac{\hbar^2}{2m}(k+g)^2 - E & \\ \ddots & & & & \ddots \end{pmatrix} = 0$$

$$g = \frac{2\pi}{a}$$

If we know $V(r)$ $\longrightarrow$

$$E_1(k), E_2(k), E_3(k), \dots$$

Free Electron, $V = 0$

$$\det \begin{pmatrix} \ddots & \dots & \dots & \dots & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & 0 & 0 & \dots \\ \dots & 0 & \frac{\hbar^2}{2m}k^2 - E & 0 & \dots \\ \dots & 0 & 0 & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & \dots & \dots & \dots & \ddots \end{pmatrix} = 0$$

$$g = \frac{2\pi}{a}$$

$$E = \frac{\hbar^2}{2m}k^2$$



$$\psi \sim \exp[ikx]$$

$$E = \frac{\hbar^2}{2m}(k-g)^2$$



$$\psi \sim \exp[i(k-g)x]$$

$$E = \frac{\hbar^2}{2m}(k+g)^2$$



$$\psi \sim \exp[i(k+g)x]$$

...

Free Electron, $V = 0$

$$\det \begin{pmatrix} \ddots & \dots & \dots & \dots & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & 0 & 0 & \dots \\ \dots & 0 & \frac{\hbar^2}{2m}k^2 - E & 0 & \dots \\ \dots & 0 & 0 & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & \dots & \dots & \dots & \ddots \end{pmatrix} = 0$$

$$g = \frac{2\pi}{a}$$

plane wave for free electrons

$$\psi \sim \exp[ikx]$$

$$\psi \sim \exp[i(k-g)x]$$

$$\psi \sim \exp[i(k+g)x]$$

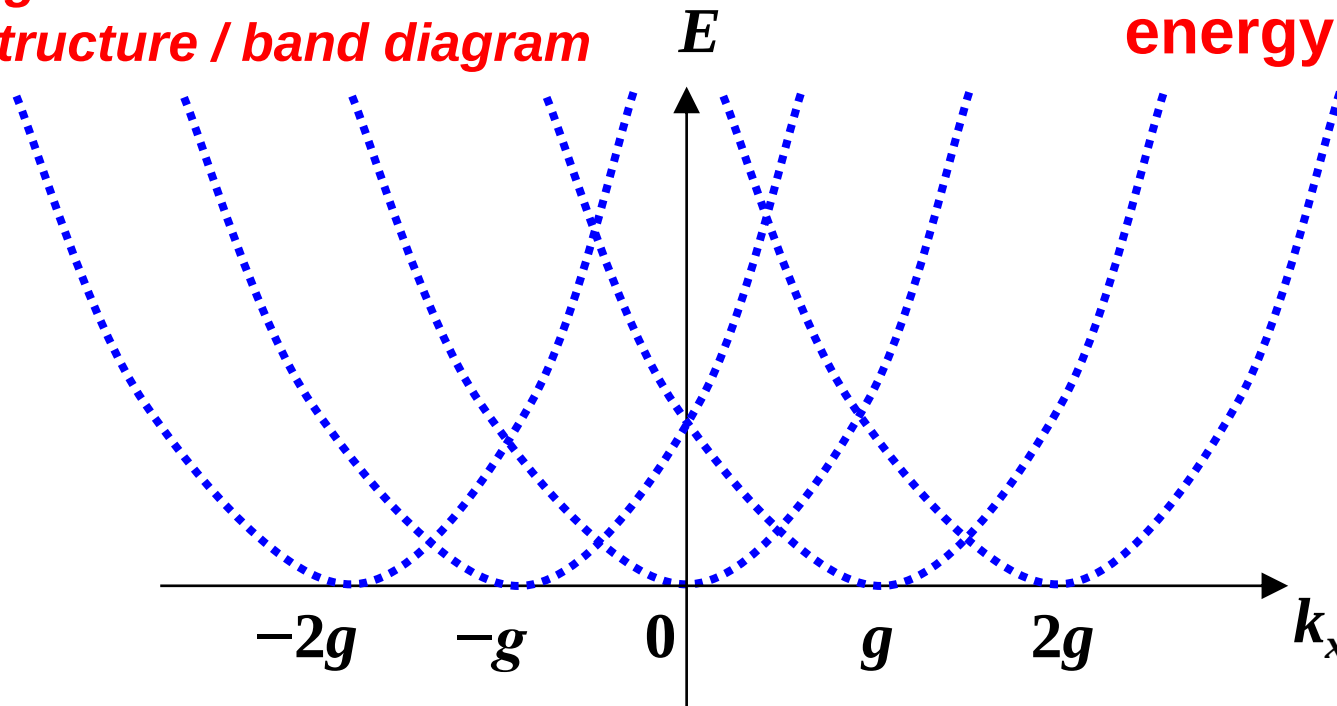
Free Electron

E-k diagram

band structure / band diagram

能带图

energy band 能带



$$g = \frac{2\pi}{a}$$

$$E = \frac{\hbar^2}{2m} k^2$$

$$E = \frac{\hbar^2}{2m} (k - g)^2$$

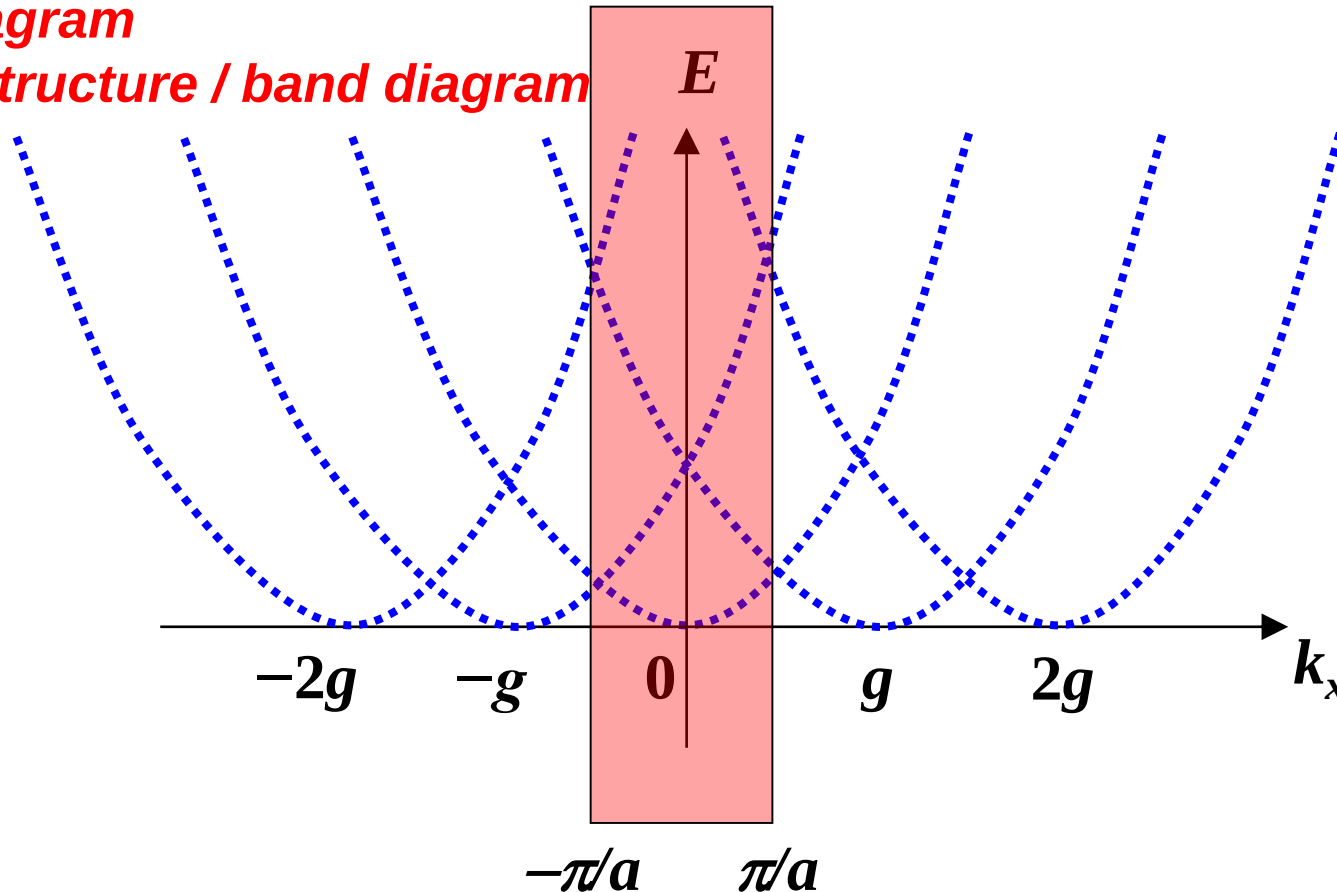
$$E = \frac{\hbar^2}{2m} (k + g)^2$$

...

$$k = 2\pi n / L \quad \text{quasi-continuous}$$

Free Electron

E-k diagram
band structure / band diagram
 能带图

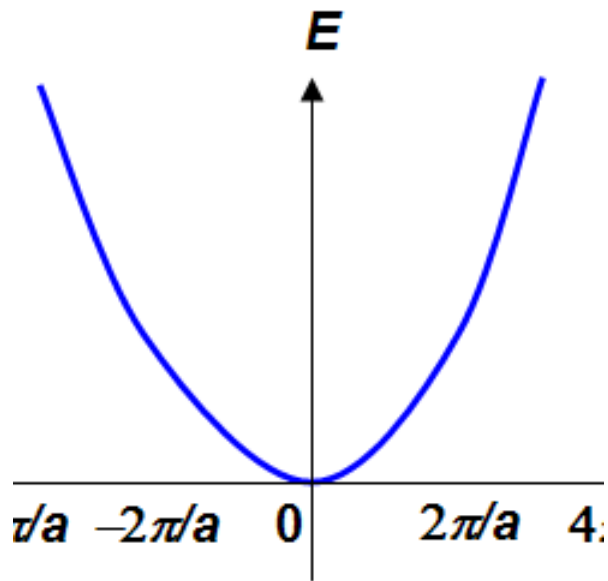


$$g = \frac{2\pi}{a}$$

***The first Brillouin zone contains
all the useful information***

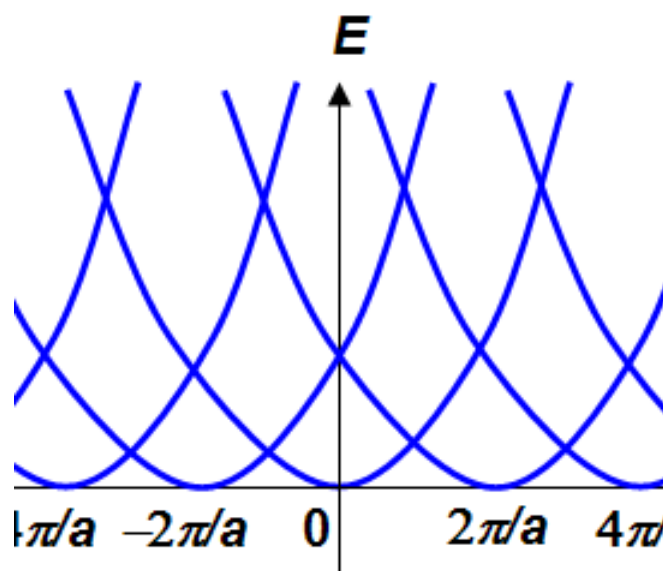
Band Structure / Diagram

Sommerfeld Model

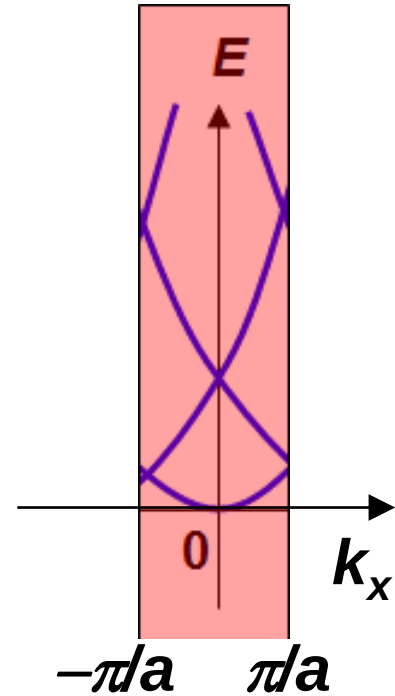


extended zone /
band structure
扩展布里渊区

+ Periodic Potentials



repeated zone /
band structure
周期性布里渊区



reduced zone /
band structure
简约布里渊区

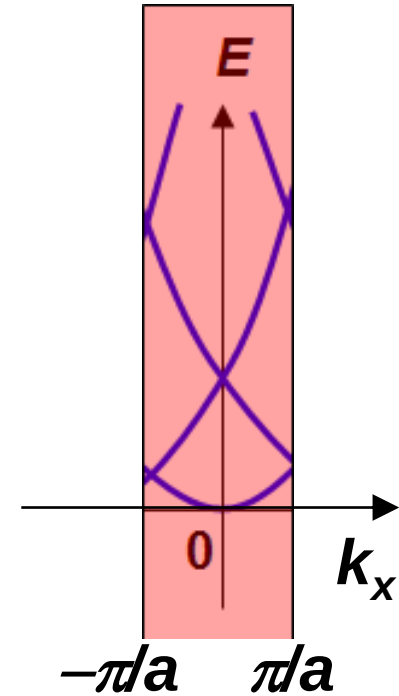
Brillouin Zones 布里渊区

- The *First Brillouin Zone (FBZ)*
 - the Wigner-Seitz cell of the reciprocal lattice
 - the volume of FBZ is a primitive cell

$$V_G = \mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)$$

- Values of $\mathbf{k}$

$$\mathbf{k} = \frac{n_x}{N_x} \mathbf{b}_1 + \frac{n_y}{N_y} \mathbf{b}_2 + \frac{n_z}{N_z} \mathbf{b}_3$$



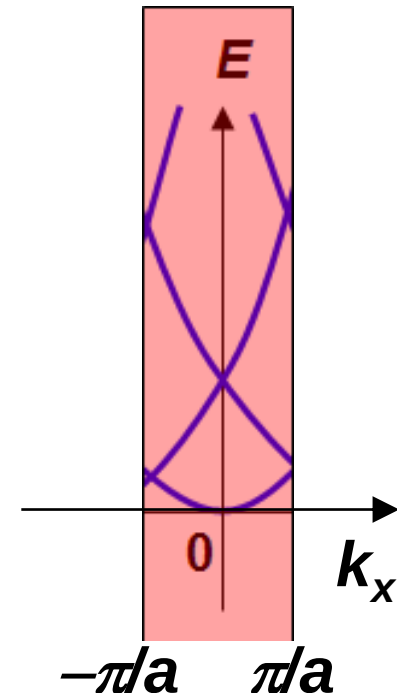
Brillouin Zones 布里渊区

- The *First Brillouin Zone (FBZ)*
 - the Wigner-Seitz cell of the reciprocal lattice
 - the volume of FBZ is a primitive cell

$$V_G = \mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)$$

- The volume of a k-state

$$V_k = \frac{\mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)}{N_1 N_2 N_3} = \frac{1}{N} V_G$$



The first Brillouin zone has N k-states, thus contains all the useful information (N - number of primitive unit cells)

An Electron in a Periodic Potential

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \cdot \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$

→

$$\left(\frac{\hbar^2 k^2}{2m} - E \right) C_{\mathbf{k}} + \sum_{\mathbf{G}} V_{\mathbf{G}} C_{\mathbf{k}-\mathbf{G}} = 0$$

The Central Equation

If we know $V(\mathbf{r})$ →

$$C_{\mathbf{k}}, C_{\mathbf{k} \pm \mathbf{G}}, C_{\mathbf{k} \pm 2\mathbf{G}} \dots$$

$$E_1, E_2, E_3 \dots$$

Bloch's Theorem

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

Ashcroft & Mermin
p.137-p.139

$$\psi(\mathbf{r} + \mathbf{R}) = \exp(i\mathbf{k} \cdot \mathbf{R}) \cdot \psi(\mathbf{r})$$



$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

$$u_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(\mathbf{r} + \mathbf{R})$$

The wave function can be expressed as
a plane wave times a periodic function

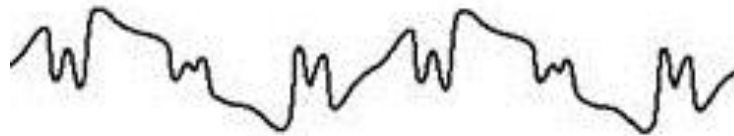
$$\exp(i\mathbf{k} \cdot \mathbf{r})$$

$$u_{\mathbf{k}}(\mathbf{r})$$

Bloch Wave

 $V(\mathbf{r})$


potential

 $\psi(\mathbf{r})$


wave function

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

Bloch wave



plane wave (free electron)



periodic function

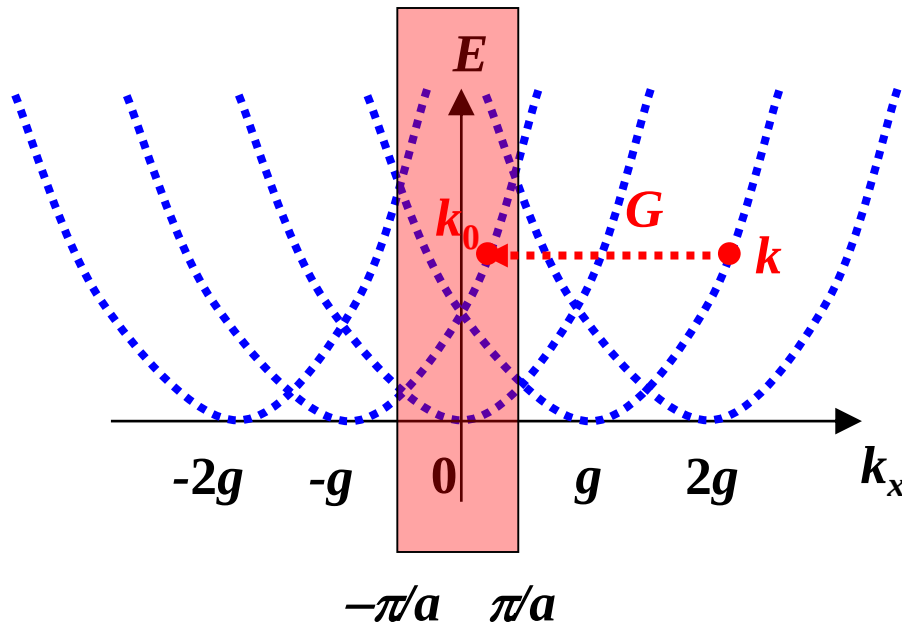
Bloch Wave

for a Bloch wave

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

it can be presented within the FBZ ($\mathbf{k} = \mathbf{k}_0 + \mathbf{G}$)

$$\psi(\mathbf{r}) = \exp(i\mathbf{k}_0 \cdot \mathbf{r}) \cdot \exp(i\mathbf{G} \cdot \mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) = \exp(i\mathbf{k}_0 \cdot \mathbf{r}) \cdot u'_{\mathbf{k}_0}(\mathbf{r})$$



$$u'_{\mathbf{k}_0}(\mathbf{r}) = \exp(i\mathbf{G} \cdot \mathbf{r}) u_{\mathbf{k}}(\mathbf{r})$$

is also periodic
because

$$u'_{\mathbf{k}_0}(\mathbf{r} + \mathbf{R}) = u'_{\mathbf{k}_0}(\mathbf{r})$$

*The FBZ contains
all the useful information*

Bloch Wave

for a Bloch wave

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

it can be presented within the FBZ ($\mathbf{k} = \mathbf{k}_0 + \mathbf{G}$)

$$\psi(\mathbf{r}) = \exp(i\mathbf{k}_0 \cdot \mathbf{r}) \cdot \exp(i\mathbf{G} \cdot \mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) = \exp(i\mathbf{k}_0 \cdot \mathbf{r}) \cdot u'_{\mathbf{k}_0}(\mathbf{r})$$

$\mathbf{k}_0$ is the **reduced vector** (简约波矢) for this wave function, which is in the FBZ

Bloch Wave

Example: for a Bloch wave in a 1D lattice (period = a)

$$\psi(x) = \sin\left(\frac{\pi}{a}x\right)$$

$$\rightarrow \psi(x+a) = \sin\left[\frac{\pi}{a}(x+a)\right] = -\sin\left(\frac{\pi}{a}x\right)$$

Bloch's Theorem

$$\psi(x+a) = \exp(ika) \cdot \psi(x)$$

$$\rightarrow \exp(ika) = -1$$



$$k = \frac{\pi}{a}(2n+1)$$

$$n = 0, \pm 1, \pm 2, \dots$$

**reduced vector
(in the FBZ)**

$$k_0 = \frac{\pi}{a}$$

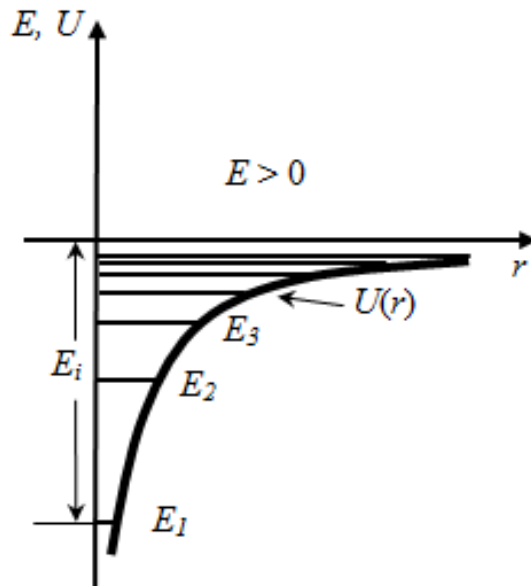
or

$$k_0 = -\frac{\pi}{a}$$

Ideal Electrons

hydrogen
atom

$$V(\mathbf{r}) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$$

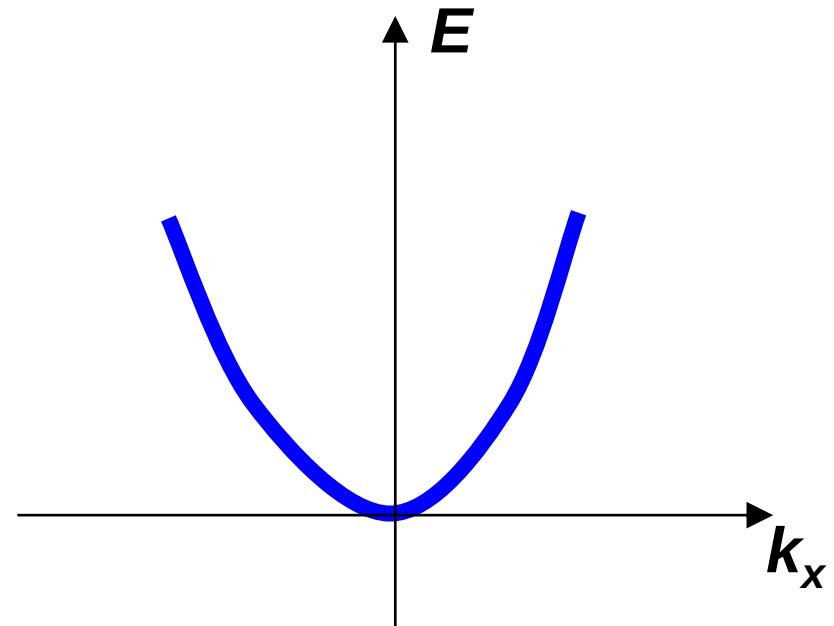


$$\psi(r, \theta, \varphi) = R_{nl}(r) \cdot Y_{lm}(\theta, \varphi)$$

purely localized

in vacuum

$$V(\mathbf{r}) = 0$$



$$\psi(\mathbf{r}) \sim \exp(i\mathbf{k} \cdot \mathbf{r})$$

purely delocalized
'free'

Real Electrons in Solids

Electrons are in *periodic* potentials

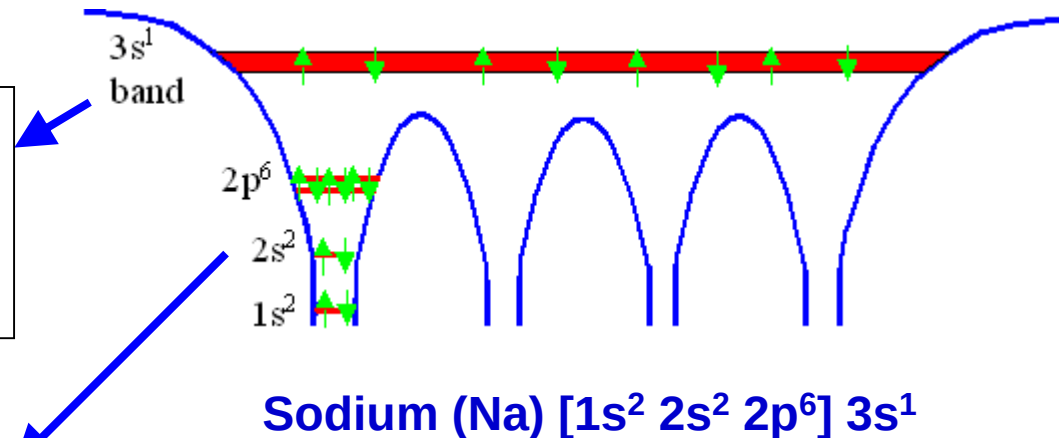


Bloch Wave

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

Nearly Free Electron Model
"近自由"近似

Tight Binding Model
"紧束缚"近似



Thank you for your attention